

module-06-moon-phases-tithi - Darshana (60-min workshop)

IKS Curriculum - middle band

Module 6 — Moon Phases and Tithi (Darśana)

Sanskrit: *Candra-pakṣa, Tithi* · **Band:** Middle (6-8) · **Duration:** 60 min · **Path:** Darśana (Workshop)

The moon's monthly cycle and the Indian lunar calendar.

Darśana note. This workshop condenses Module 6's full 10-day arc into one hour. For the deeper version — daily lesson plans, quizzes, assessments, slide decks — switch to [Adhyayana](#) (the Full Course path).

Opening hook · 5 min

Project a year-long animation of moon phases. “~29.5 days from full moon to full moon. Indian astronomy divides this into 30 lunar days called tithis. Why 30 and not 29 or 30 sometimes? Today we find out.”

Core idea · 15 min

Moon-month framework:

- **Synodic month:** ~29.5 days from one full moon to the next.
- **Tithi:** 1/30th of synodic month — NOT a 24-hour day. Tithi length varies (~20-26 hours).
- **Pakṣa:** half-month, 15 tithis. *Śukla* (bright/waxing) or *Kṛṣṇa* (dark/waning).
- **Pañcāṅga:** the five-limbed Indian calendar with tithi, vāra (weekday), nakṣatra, yoga, karaṇa.

Why eclipses? Solar eclipse only on amāvasyā (new moon). Lunar eclipse only on pūrṇimā (full moon). Both require alignment of Sun-Moon-Earth.

Demonstration · 10 min

Live computation: pick today's date, look up the lunar elongation (in degrees) from any pañcāṅga app. Compute the tithi by dividing by 12°. Compare with the published pañcāṅga.

Source moment · 5 min

Paraphrase Sūrya-siddhānta, ch. 2: *“The tithi is the time required for the Moon to gain 12° of longitudinal separation from the Sun. There are 30 tithis in a synodic month.”*

Hands-on activity · 15 min

Build a class tithi calendar for the current month. Each student fills in 2-3 dates with tithi, pakṣa, and one cultural festival or saint’s day if any.

Reflection + take-home · 10 min

“Why does India use two calendars (lunar pañcāṅga + Gregorian solar) in daily life? What does each handle better?”

Go deeper into Adhyayana

The 10 days you’re skipping if you only do this workshop:

- **Day 01** — The moon’s shape — why it changes
- **Day 02** — Pakṣa — bright and dark fortnights
- **Day 03** — 15 tithis — names and computation
- **Day 04** — Building a tithi calendar
- **Day 05** — Lunar months and festivals
- **Day 06** — Pañcāṅga — the five limbs
- **Day 07** — Eclipse geometry
- **Day 08** — Modern lunar science — phases, libration, tides
- **Day 09** — Cross-cultural calendars — Hijri, Chinese, Hebrew
- **Day 10** — Synthesis quiz + moon diary share

→ [Open the Adhyayana track for this module](#) to access all 10 days, the 4-audience PDFs, the slide deck, the quizzes, and the project rubric.